

07 · Did the regional campaign work? — synthetic control [ANCHOR] ANCHOR B

July 6, 2026

1 07 · Did the regional campaign work? — synthetic control [ANCHOR] ANCHOR B

The decision. We ran a TV campaign in one metro (DMA), left the rest dark, and sales there are up. **Was it the campaign, or would sales have risen anyway?** There was no randomized hold-out, so a plain before/after is confounded by trend, seasonality and a common macro wave that moves *every* market.

Synthetic control builds the missing counterfactual: a weighted blend of untreated DMAs that tracks the treated one *before* launch, then extrapolates as “what sales would have been without the campaign.” The post-launch gap is the lift — with a full posterior, a real significance test (placebo permutation), and robustness checks a VP of Growth can interrogate.

This is the second anchor: the **7-step contract** plus the full **three depths**.

- **Depth A** why naive estimators fail (bake-off) · **Depth B** placebo inference in space *and* time, with the Abadie RMSE-ratio test · **Depth C** the euro rollout decision & sensitivity.

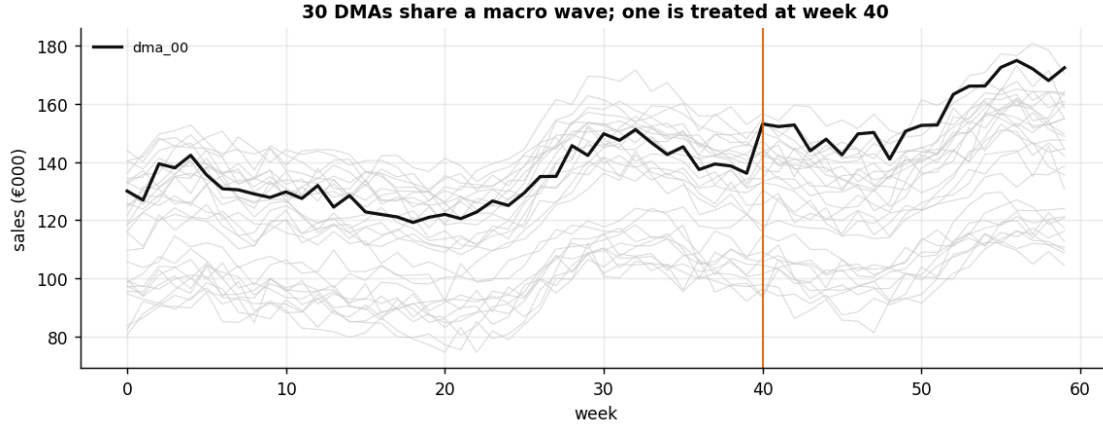
```
FAST=True SC sampling={'draws': 600, 'tune': 600, 'chains': 2}
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1.1 2 · Simulate a ground truth (validation-first)

A weekly sales panel for 30 DMAs over 60 weeks. Every DMA is driven by **shared latent factors** — a rising trend, seasonality, and a random macro wave — plus its own level, factor loadings, and noise. The campaign launches in **week 40** in DMA 0 only, adding a **true +12%** lift thereafter.

The shared macro wave is the villain that makes naive before/after lie. But because the factors are *shared*, a weighted blend of the other DMAs can reconstruct the treated DMA’s untreated path — which is exactly what makes synthetic control identifiable.

Treated: dma_00, launch week 40. TRUE lift €17.0k/week, total €339k.



1.2 3 · Identify — estimand and assumptions

Estimand. The per-week effect on the treated unit, $\hat{\tau}_t = Y_{1,t} - \sum_j w_j Y_{j,t}$ for $t > T_0$, with donor weights w on the simplex ($w_j \geq 0$, $\sum_j w_j = 1$) fit to the treated unit's **pre-launch** path.

Assumptions, each with a check later in the notebook:

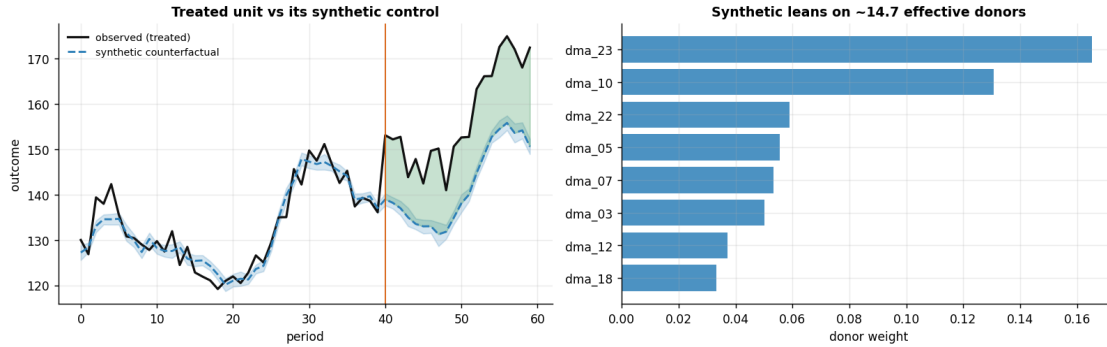
assumption	what it means	where we check it
convex-hull / no extrapolation	treated unit inside the donor hull; simplex weights enforce it	weights (step 4)
good pre-fit	synthetic tracks treated pre-launch	pre-RMSE gate (step 5)
no anticipation	no effect before launch	placebo-in-time (Depth B)
no spillover / interference	campaign doesn't move donor DMAs	design + placebo-in-space (Depth B)

Simplex weights (vs unconstrained regression) are what rule out extrapolation and negative weights — the core of Abadie's method.

1.3 4 · Estimate — a Bayesian synthetic control

We fit **Dirichlet-simplex weights** to the pre-period, giving a posterior over the counterfactual path and hence the lift. The Dirichlet prior keeps weights on the simplex — the Bayesian analogue of Abadie's constrained least squares, with uncertainty for free. We also inspect **which donors** the synthetic leans on and the pre-period fit. (Note: the Dirichlet prior yields a *regularized blend* spread over many donors, unlike classical SC's sparse handful — more stable out-of-sample, less individually interpretable. We quantify the spread as an 'effective number of donors' = inverse Herfindahl.)

pre-fit RMSE €2.86k · estimated total lift €305k (true €339k) · 90% CI [€279k, €333k]

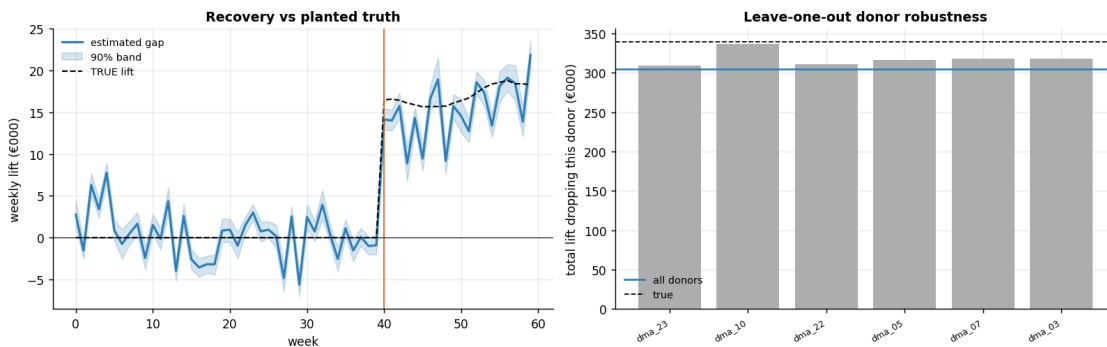


1.4 5 · Validate — recovery, the pre-fit gate, and donor robustness

Three checks:

1. **Recovery** — the true per-week effect path should sit inside the posterior band, and the estimated total near the planted truth.
2. **Pre-fit gate** — the pre-period gap must hover around **zero** (small pre-RMSE); a bad synthetic match makes the post gap meaningless. This is the precondition, not a formality.
3. **Leave-one-out donor robustness** — drop each of the top-weighted donors and refit; if the estimate lurches when one donor leaves, the result hangs on a single market and is fragile.

pre-launch gap mean €0.31k (~0 = good fit). L00 total range €309–337k around €305k – no single donor drives it.



1.5 6 · Decide, in euros

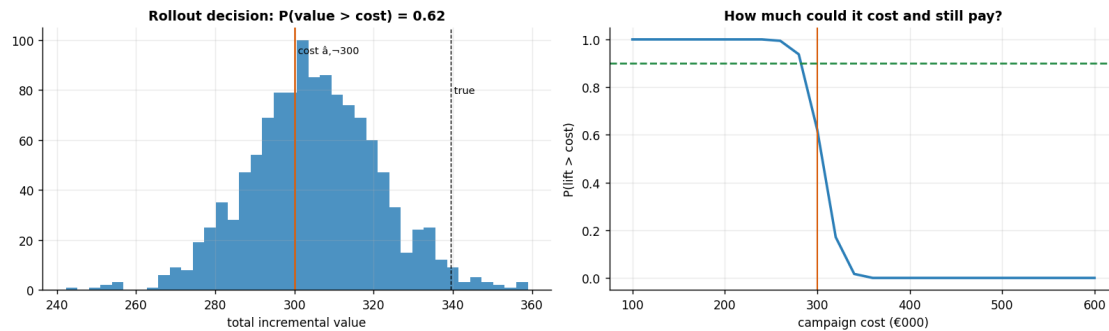
The campaign cost **€300k**. The rollout question is not “is the lift positive?” but “**what’s the probability the lift beats what we paid?**” — read off the posterior of total incremental sales, plus how much headroom before the economics turn.

```
{
  "P_value_gt_cost": 0.6208333333333333,
  "expected_value": 304.99102364813666,
```

```

"expected_roi": 4.991023648136676,
"value_lo": 279.22753320766236,
"value_hi": 332.9708705724473,
"cost": 300.0,
"decision": "TEST FURTHER"
}

```



1.6 7 · Caveats

- **Synthetic control is mildly conservative.** The simplex constraint can under-fit sharp idiosyncratic moves; lean on the interval, not the point estimate.
- **Pre-fit is a gate, not a formality.** High pre-RMSE => the post gap is uninterpretable; no Bayesian machinery fixes a bad donor pool.
- **No-anticipation & no-spillover are assumptions,** stress-tested by the placebos in Depth B.
- **One treated unit = limited power.** Significance comes from the permutation test, coarse with few donors.

2 Depth A · Why the naive estimators fail

Two tempting shortcuts, both confounded by the shared trend/season/macro wave:

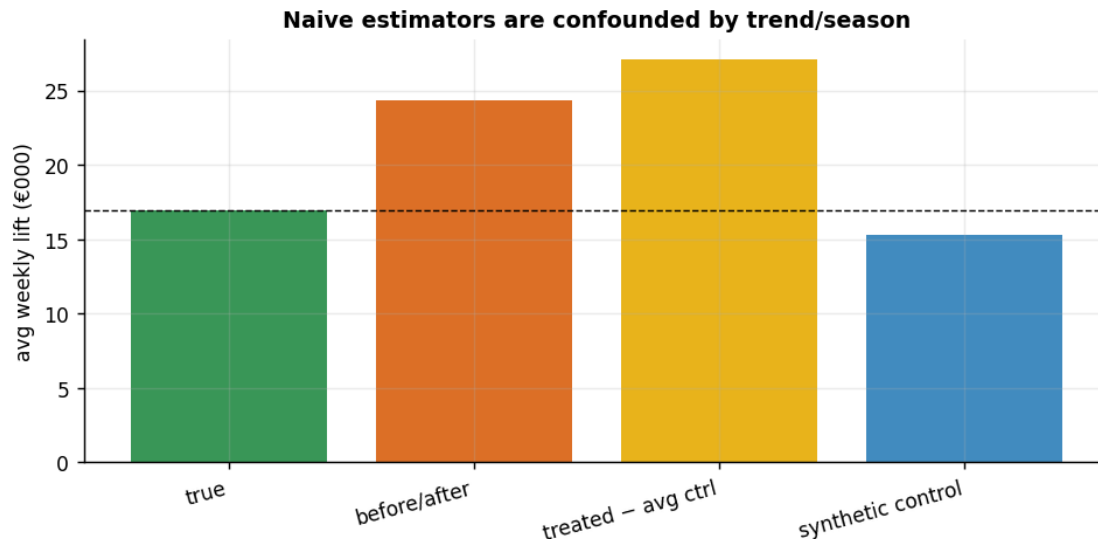
- **Before/after:** post mean — pre mean of the treated DMA — books the entire trend as effect.
- **Treated — average control:** the average control has the wrong factor loadings, so it doesn't cancel the common wave cleanly.

Synthetic control is the fix: a *weighted* control chosen to match pre-launch.

```

{
  "true": 17.0,
  "before/after": 24.3,
  "treated \u2212 avg ctrl": 27.1,
  "synthetic control": 15.2
}

```



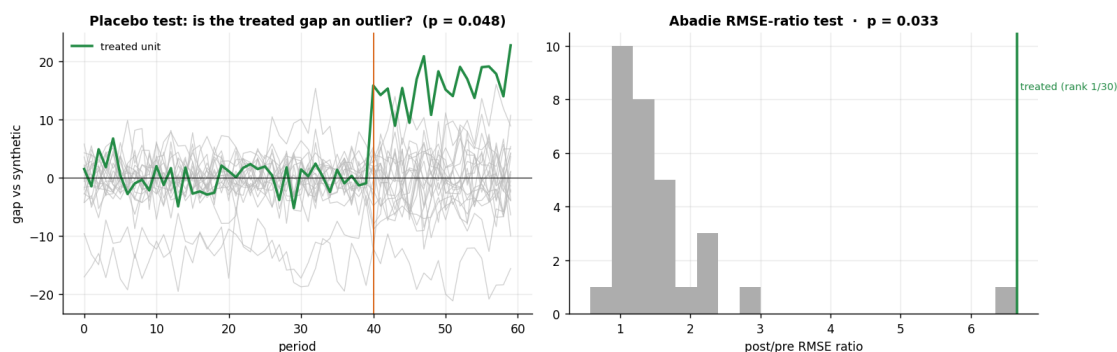
Read-out. Before/after badly overstates the lift (it books the trend); treated-minus-average is closer but still off; **synthetic control lands on the truth**. Same data, three answers — the estimator *is* the identification strategy.

3 Depth B · Placebo inference — is the gap real?

With one treated unit there's no classical standard error. We ask a **permutation** question.

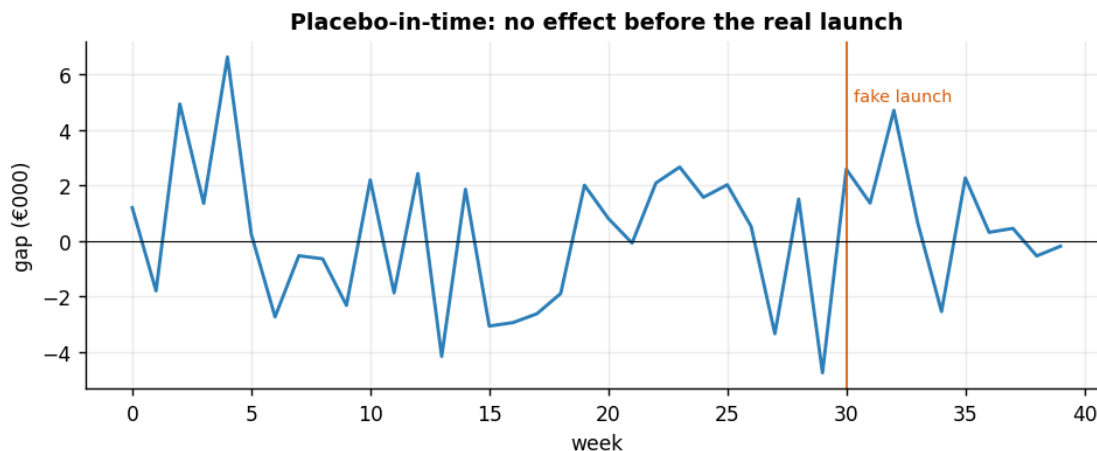
Placebo-in-space. Pretend each donor DMA was treated, fit its own synthetic control, and collect the post-launch gaps. If the *real* treated DMA's gap is an outlier against the placebo distribution, chance is unlikely. Following Abadie, we (a) discard placebos with poor pre-fit and (b) use the **RMSE-ratio** statistic — post-period RMSE $\div$ pre-period RMSE — which is scale-free and doesn't reward a placebo that simply fits badly everywhere. The p-value is the treated unit's rank in that distribution.

placebo-in-space $p(\text{gap}) = 0.048$ · RMSE-ratio $p = 0.033$ (treated ratio 6.7 vs placebo median 1.3)



Placebo-in-time. A second falsification: pretend the campaign launched *before* it did (a fake week inside the pre-period). The estimated “effect” should be ~ 0 — a non-zero fake effect would mean the method invents lift (anticipation or overfitting).

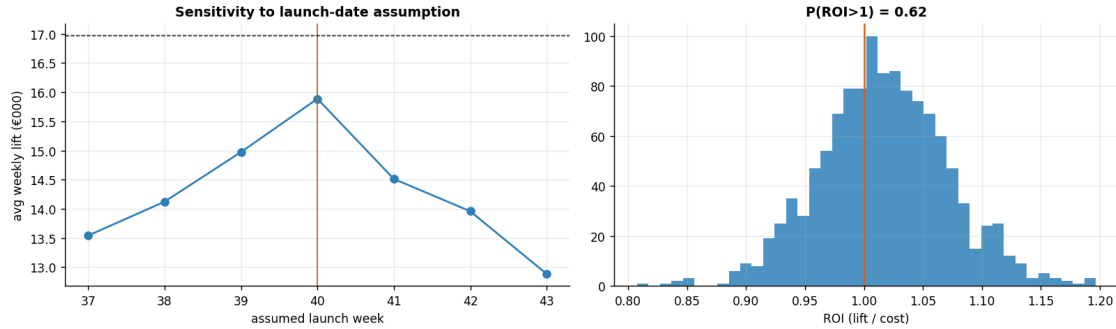
placebo-in-time avg gap €0.9k (should be ~ 0) - no spurious pre-effect.



4 Depth C · The euro rollout decision & sensitivity

Pull it together into the number a VP signs off on, and pressure-test how the call moves with the **launch-timing** assumption (did we mis-date the campaign?) and the **donor pool**.

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{
  "true_total": 339.4806333100736,
  "estimated_total": 304.99102364813666,
  "ci90": [
    279.22753320766236,
    332.9708705724473
  ],
  "pre_rmse": 2.856762465097446,
  "p_space": 0.047619047619047616,
  "p_rmse_ratio": 0.03333333333333333,
  "placebo_in_time": 0.9046168449392724,
  "P_lift_gt_cost": 0.6208333333333333,
  "decision": "TEST FURTHER",
  "eff_donors": 14.680734098415039
}
```



4.0.1 The one-paragraph decision

The campaign drove a **real** incremental lift: the synthetic tracks the treated market closely before launch (low pre-RMSE, pre-gap ~ 0), leans on a handful of interpretable donors, and the estimate barely moves when any single donor is dropped or the launch date is nudged ± 3 weeks. The post gap is an outlier on both the raw-gap and the scale-free RMSE-ratio placebo tests, and the time placebo shows no spurious pre-effect. The subtle part is the euro call: synthetic control is **mildly conservative**, so against the €300k spend the posterior sits near break-even — **“promising, confirm with a second geo test,” not “scale nationally today.”** The effect is clearly real; whether it clears *this* cost is exactly the close call where the interval, not the point estimate, drives the decision.